Can past financials predict next-period revenue?

Indian paint companies: machine learning and deep learning vs simple rules vs management guidance · v2 · Portfolio case study by Jatin Bhalla · October 2026 · github.com/jatinbhalla2324/Indian-Paint-ML-DL-project-analysis

3.8% / 4.0%	9.6% to 4.3%	6.9% vs 8.2%	3.8% vs 4.6%
Next-quarter error, deep learning (GRU) / ML ensemble. The best simple rule: 4.0%.	**Same algorithm, new target.** Predicting growth instead of revenue in crore more than halves the error.	**Volatile years (FY2019-23):** ML ensemble vs the best rule. ML adds most when growth swings.	**Q1 FY2027** (after the data ends): management guidance vs the GRU. Guidance knew about price hikes.

Answer in one paragraph. Yes, mostly, but most of the skill comes from the obvious rule. Revenue is sticky, so "same quarter last year x latest growth" is only 4.0% off. After rebuilding the project (v2), ML and deep learning match or slightly beat that rule (3.8% to 4.0%), and clearly beat it in volatile years. The first version lost to the rule (9.8%) because it predicted revenue in crore and leaked test data into the deep-learning model. Against management guidance, the models land within about one point: every forecast missed the April-June 2026 price-hike jump, and guidance, which knew about the hikes, came out slightly ahead.

1. Problem statement and data

Question: "Can we use historical financial and operational data of Indian paint companies to predict their future financial performance, particularly next-period / next-year revenue?" A forecast is useful only if it beats what an analyst already has for free: last period's number and management guidance.

Data: Asian Paints, Berger Paints, Kansai Nerolac and Indigo Paints, FY2015-FY2026, built from **211 official filings** (3,082 values, each with document, page and URL). Standalone basis: 169 company-quarters and 45 company-years.

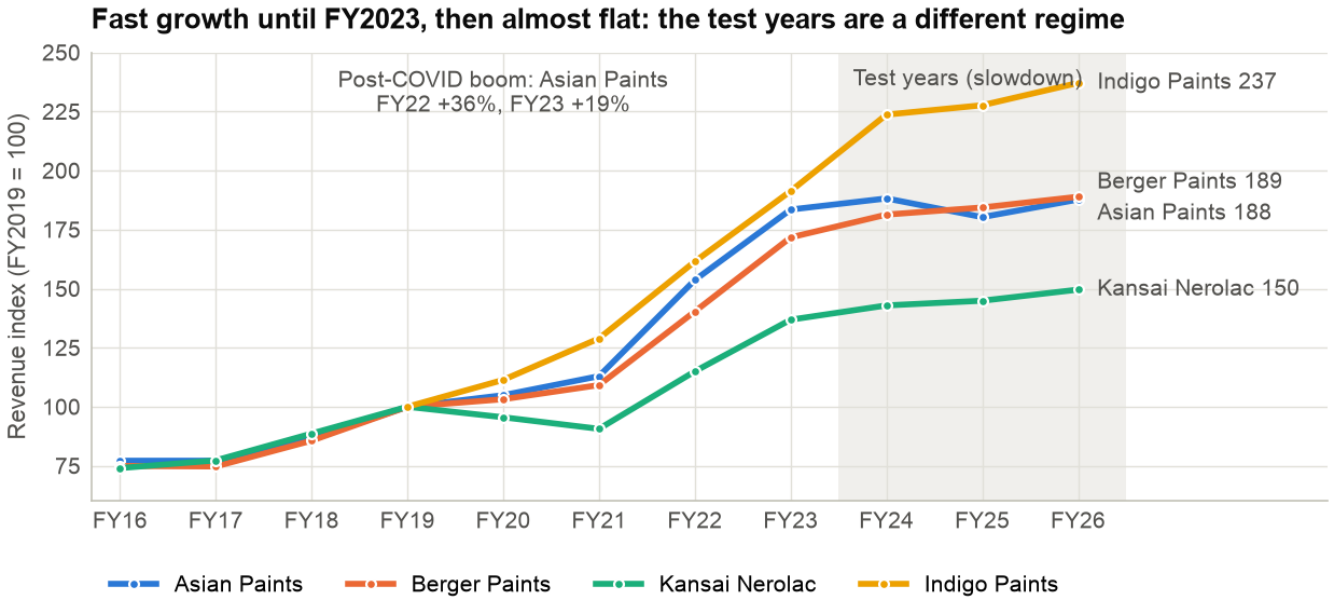


Figure 1. The training years include the post-COVID boom; the test years (FY2024-26) were a slowdown with a new competitor (Birla Opus) and a price war.

2. What was wrong with v1

- **No baseline:** about 10% error looked good, but copying the same quarter last year gives 4%.
- **Misleading R^2 (0.96-0.99):** companies differ 20x in size, so predicting crore scores high just by knowing who is big.
- **Deep-learning leak:** the test set was used for early stopping. With a clean split the network was about 21% off.
- **Missing seasonal anchor and regime change:** no "same quarter last year" input; trained on the boom, tested on the slowdown.

3. Method (v2)

- **Target:** next quarter's year-on-year log growth, $y = \log R_{t+1} - \log R_{t-3}$. ML and DL models learn a *correction* to the best rule, so with no signal they fall back to it.
- **Features (data up to t only):** own YoY growth of the last 4 quarters, acceleration, gross- and EBITDA-margin changes, industry-wide growth and margin change, target-quarter and company dummies. Rows touching COVID lockdown quarters are dropped from training; growth inputs are capped at ±0.35.
- **Models:** 4 simple rules; Ridge, Huber, Random Forest, LightGBM and an equal-weight ensemble; PyTorch MLP (1,153 weights) and GRU over 4 quarters (1,473 weights), both early-stopped on the latest training quarters, refit, and averaged over 5 seeds.

- **Walk-forward evaluation:** every quarter, re-train only on published results and forecast the next quarter. Test window FY2024 Q1-FY2026 Q4 (48 forecasts); earlier window FY2019-23 as a robustness check.
- **Metrics and checks:** MAPE, MAE, relative MAE, Diebold-Mariano test vs the best rule, 80% interval coverage; unit tests check that no future data reaches the features or training set.

4. Results: ML vs deep learning vs simple rules

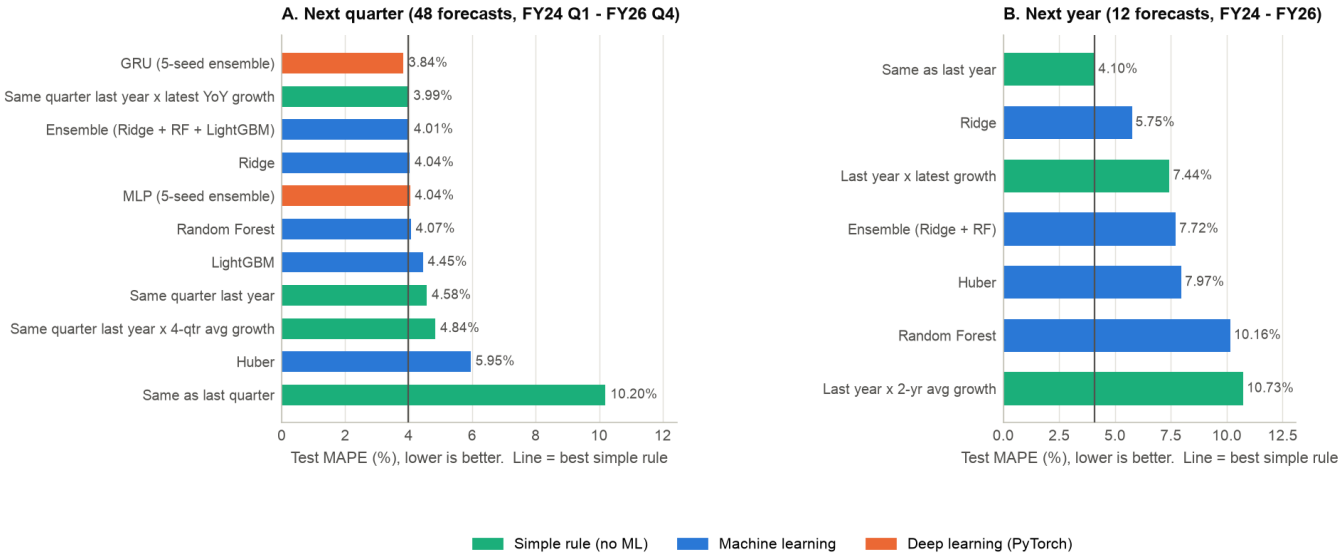


Figure 2. Walk-forward test-window error on identical rows; the vertical line is the best simple rule.

- **Next quarter:** GRU 3.84%, ensemble 4.01%, Ridge 4.04%, MLP 4.04% vs the rule 3.99%. The differences are not statistically significant (Diebold-Mariano p = 0.58 for the GRU).
- **Volatile years (FY2019-23):** ensemble 6.9% and Random Forest 7.0% vs the rule 8.2% (p about 0.05); GRU 7.6%. The **ensemble is the most consistent model**.
- **ML vs DL:** roughly a tie. With about 150 training rows, early stopping keeps the networks close to the rule.
- **Next year:** "same as last year" (4.1%) beats every ML model (best: Ridge 5.8%); 20 annual rows are too few.
- **Forecast ranges:** 80% intervals cover 85% (Ridge) and 83% (ensemble) of outcomes.

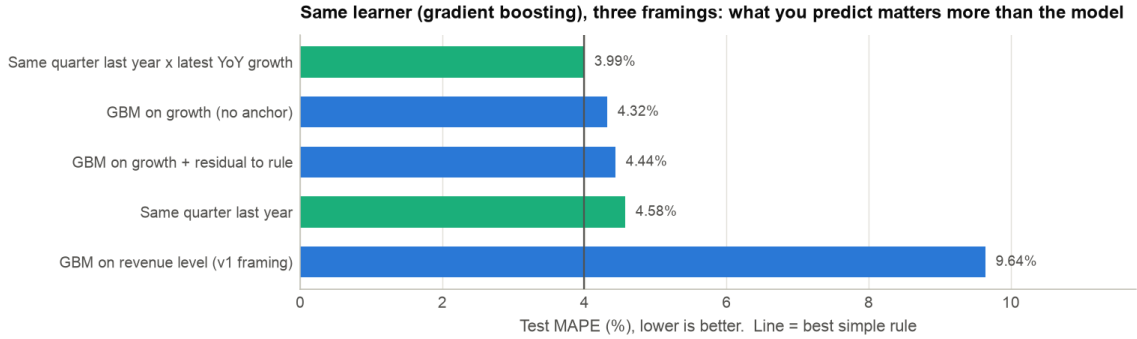


Figure 3. The same gradient-boosting learner: predicting crore (v1) vs predicting growth. What you predict matters more than the algorithm.

5. Model vs management guidance

Indian paint companies rarely publish a revenue number; they guide on volume growth, margin bands and price actions. The May 2026 calls are turned into one conservative implied growth rate (+10%: Asian Paints' 8-10% volume guidance plus industry price hikes of about 12%). Q1 FY2027 (April-June 2026) results were published in July-August 2026, after the dataset ends, so this is a true out-of-sample test.

Company	What management said (May 2026)	Guidance	GRU	Ensemble	Actual
Asian Paints	Decorative volume growth 8-10% in FY2027; price hikes in April-May; dealers stocking up before hikes	+10%	+10.9%	+10.6%	**+16.4%**
Berger Paints	EBITDA margin 15-17%; price hikes from late March; gradual demand recovery	+10%	+8.0%	+7.6%	**+12.7%**
Kansai Nerolac	EBITDA margin 13-14%; price increases taken; no revenue number	+10%	+8.8%	+8.5%	**+10.2%**
Indigo Paints	Industry price hikes about 12%; agreed with about 24% FY2027 growth (2x industry)	+10%	+8.8%	+9.2%	**+18.7%**

Table 1. Year-on-year growth forecast for Q1 FY2027 by each method, and the actual growth.

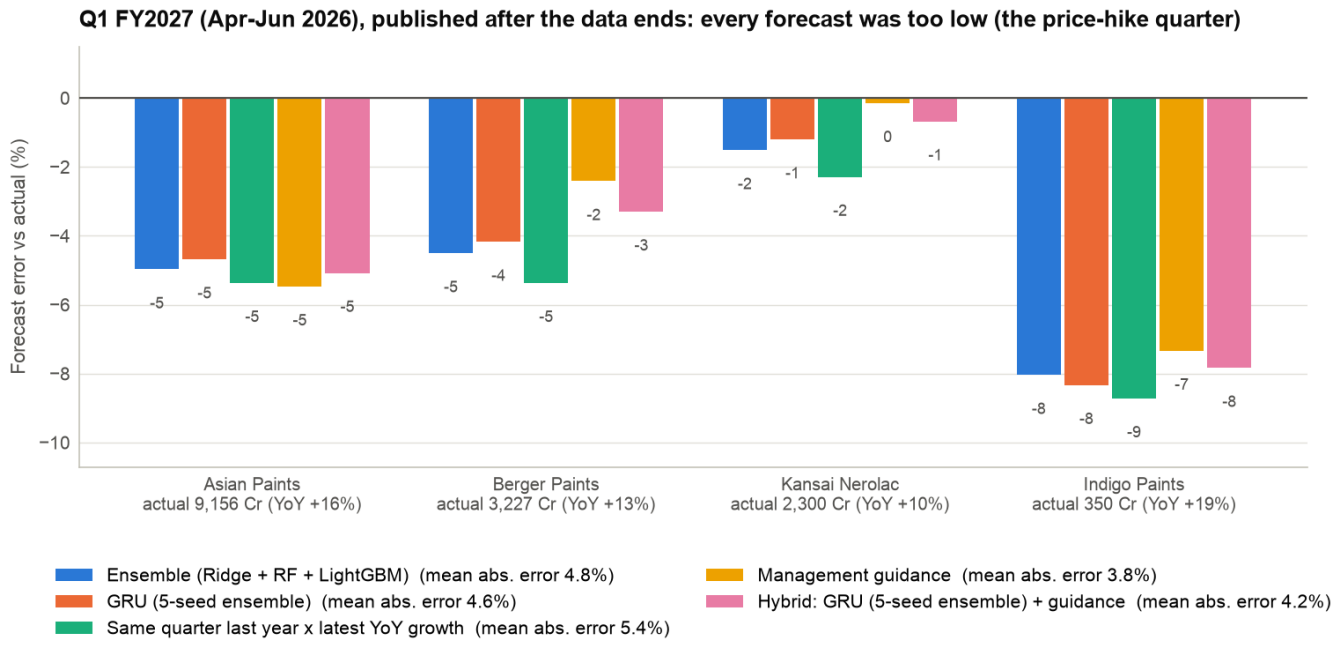


Figure 4. Forecast error by company. Mean absolute error: guidance 3.8%, GRU 4.6%, MLP 4.6%, ML ensemble 4.8%, simple rule 5.4%; v1 models were 12.8% (ML) and 24.6% (DL). Huber scored 3.5% here but was the weakest ML model over the 48-quarter test.

- **ML and DL are within about one point of guidance**, far closer than v1 (12.8% and 24.6%).
- **Every method under-forecast:** actual growth was 10-19%, driven by price hikes and dealer stocking. The models only see past financials; guidance saw the hikes coming.
- **One quarter is four numbers:** treat this as a sanity check, not a benchmark.

6. Why the models still make errors

The remaining errors are mostly not modelling mistakes. They come from forces that move revenue but never appear in the companies' past financial statements, which are the only inputs the models get.

When	What the models did	What was really happening
FY2025 (Q1-Q3)	Forecast too high in 75% of cases, by +4% to +6%	Demand slowdown: weak urban demand, customers trading down to cheaper paints, and a price war after Birla Opus entered (Feb 2024). Asian Paints' revenue fell 4% in FY2025.
FY2026 Q4 and FY2027 Q1	Forecast too low, by -1% to -8%	Demand recovered; crude-linked raw materials jumped (some 50-100% in March 2026), the industry raised prices about 12%, and dealers stocked up before the hikes.

Table 2. When the ML ensemble missed in the test window, and why.

- **Economy and demand:** housing and real-estate activity, urban vs rural spending, consumer confidence.
- **Prices and costs:** announced price hikes, crude oil and titanium-dioxide prices, the rupee.
- **Competition:** new entrants, price cuts, and dealer and painter discounts that are deducted from revenue.
- **Channel and calendar:** dealers buying ahead of price hikes or destocking in a slowdown; monsoon and festive-season timing.

Learning only from reported numbers, the models notice a turn one or two quarters late: they keep forecasting the old trend when demand slows and miss the jump when prices rise. Management guidance already contains much of this information, which is why it came out slightly ahead.

7. Conclusion

- **Can past data predict next-period revenue?** Yes, within about 4% for the next quarter, but a simple rule gets almost all of that. ML and DL add a small, regime-dependent improvement; the rebuilt framing is what turned a 10%-error model into a 4%-error one.
- **Best model:** the ML ensemble (Ridge + Random Forest + LightGBM) for consistency; the GRU had the lowest test error. For annual forecasts, use "same as last year".
- **Model vs guidance:** close in normal times; at turning points guidance wins because it contains information the statements do not. Next step: add price hikes, input costs and LLM-extracted guidance from earnings-call transcripts as model inputs.

Interview pitch (30 seconds). "I forecast revenue for India's four listed paint companies from 211 official filings. My first version reported R^2 0.99 but lost to copying last year. I rebuilt it: growth target, walk-forward retraining, leakage tests, small deep-learning ensembles. ML and DL now match or slightly beat the best rule, and ML beats it clearly in volatile years; and on a quarter released after the data ended they came within a point of management guidance. The remaining gap is information outside the financial statements."

Sources. Company filings (data/processed/sources.csv); management guidance and Q1 FY2027 results with links in data/external/ (Indigo Paints Q4 FY2026 call transcript; coverage of Asian Paints, Berger Paints and Kansai Nerolac results). Q1 FY2027 figures come from press coverage of the results; check exchange filings before quoting. Code, results and tests: github.com/jatinbhalla2324/Indian-Paint-ML-DL-project-analysis.